

Advanced Data Analysis Documentation

AnalystLab Africa – Data Analytics Internship | Week 3

Dataset: Superstore Sales Dataset

This document records the analytical process used to extend the Week 2 Superstore project into a more advanced Week 3 business analysis. The analysis was implemented in Power BI using calculated measures, interactive visuals and business-focused questions.

1. Analysis Objective

The objective was to move beyond descriptive dashboard reporting and investigate what is driving sales and profitability, where performance is weak, how discounting affects margins, which products and segments matter most, and which areas require management action.

2. Advanced Business Questions

- What is driving overall sales and profit performance?
- Which categories generate strong revenue and strong profit?
- Which products generate high sales but weak profit or losses?
- How large is the loss-making product portfolio?
- Which regions provide the strongest performance and which require intervention?
- Which customer segment contributes the most revenue and profit, and how healthy is its margin?
- Are high discounts concentrated in specific categories or sub-categories?
- How are sales and profit changing over time?

3. Analytical Areas Covered

Sales Performance

Total sales, sales trend, annual performance and sales growth.

Profitability Analysis

Total profit, profit margin, profit growth and category/regional profitability.

Product Performance

Top products, low-profit/loss-making products, sales-versus-profit relationships and SKU-level profitability.

Regional Performance

Sales and profit by region, regional comparison and identification of underperforming regions.

Customer & Segment Analysis

Sales and profit contribution by customer segment and margin evaluation.

Discount Analysis

Average discount patterns, especially within Furniture and its sub-categories.

4. KPI and Calculated Measures

- Total Sales – total revenue generated from transactions.
- Total Profit – total profit generated from transactions.
- Total Orders – total order volume used to assess business activity.
- Average Sales – average sales value used to understand typical transaction performance.
- Profit Margin – profit expressed relative to sales to evaluate profitability quality.
- Sales Growth – year-over-year change in sales.
- Profit Growth – year-over-year change in profit.
- Loss-Making Product – identifies/counts products with negative profitability.

5. Key Analytical Findings

Category performance

Technology leads with approximately \$0.84M revenue and \$145K profit. Furniture has approximately \$0.74M revenue but only \$18K profit.

Discount concentration

Tables show about 24% average discount and Bookcases about 21%, making them major discount-management priorities.

Product profitability

299 of approximately 1.85K products are loss-making, equivalent to about 16% of the product portfolio.

Regional performance

West is the top region at approximately \$0.73M revenue and \$108K profit. Central has the lowest regional profit at about \$40K and includes a Furniture loss of about \$3K.

Customer segment

Consumer contributes approximately \$1.2M revenue and \$134K profit, but its margin is about 11.5%, making margin improvement in this large segment commercially important.

6. Business Problem Investigation

Problem 1 – High sales but weak profit

Furniture demonstrates that strong revenue does not automatically mean strong financial performance. The analysis therefore focuses on discounting, product mix and costs.

Problem 2 – Loss-making products

The 299 loss-making products represent a portfolio-level profitability issue requiring SKU-level review.

Problem 3 – Regional underperformance

Central's low profit, including Furniture losses, indicates that regional revenue should be evaluated together with profit.

Problem 4 – Discount-driven margin pressure

High average discounts in Tables and Bookcases indicate areas where pricing discipline can be strengthened.

Problem 5 – Large segment with modest margin

Consumer is the largest segment but its approximately 11.5% margin indicates an opportunity to improve profitability without needing to grow the customer base first.

7. Analytical Approach

- Reviewed Week 2 findings and identified unanswered profitability, product, regional, customer and discount questions.
- Used Power BI measures to calculate core KPIs and growth metrics.
- Compared sales and profit rather than relying on sales alone.
- Used category, sub-category, product, region and segment breakdowns to isolate performance differences.
- Used discount analysis to identify areas where pricing pressure may be contributing to weak margins.
- Used interactive slicers for Year, Region, Category and Segment to test the analysis from different business perspectives.
- Converted analytical results into business problems, insights, risks, opportunities and recommendations.

8. Assumptions

- Profitability is assessed using the Profit field supplied in the Superstore dataset.
- Profit Margin is interpreted as profit relative to sales.
- Loss-making products are products with negative profit under the dashboard's loss-making-product measure.
- Year-over-year growth compares the selected period with the corresponding prior period.
- Discount analysis is treated as an indicator of pricing pressure; it does not by itself prove that discounting caused every observed profit outcome.

9. Limitations

- The analysis is based on historical transaction data and does not include external factors such as competitor pricing, inflation or market conditions.
- Observed relationships between discount and profit should not automatically be interpreted as causal relationships.
- The dataset does not provide all cost components needed for a full contribution-margin or product lifecycle assessment.

- Recommendations should be validated with operational, pricing, inventory and customer-level information before implementation.

10. Advanced Analysis Conclusion

The advanced analysis demonstrates that the business should focus on profitable growth rather than revenue growth alone. Technology provides a strong platform for expansion, while Furniture discounting, loss-making products and Central-region profitability require targeted intervention. The analysis therefore converts the Week 2 descriptive findings into measurable management actions supported by KPIs, product-level investigation, regional comparison, segment analysis and discount evaluation.